

AC2205 Marking Scheme

Intermediate Management Accounting · Summer 2023 · Paper 1

Examination Session and Year	Summer 2023
Module Code	AC2205
Module Title	Intermediate Management Accounting
Paper Number	1
External Examiner	Prof Martin Quinn
Head of Department	Prof Mark Mulcahy
Internal Examiner	Dr Peter Cleary
Duration of Paper	1.5 hours
Special Requirements	Students are allowed to use calculators that have standard arithmetic, scientific, graphical or algebraic functions provided that they are NOT programmable and can NOT store text (other than algebraic notation). Show all your workings.

Instructions to Candidates (as per exam paper): Answer two questions only. All questions carry equal marks (50 each). Show all your workings.

Own-figure rule: where a candidate makes an error at an early stage of a multi-step numerical question, full method marks are awarded for correctly applying the required technique to their own (incorrect) figure in all subsequent steps. Marks are never cascaded away for a single early error.

Marking philosophy: this is a closed-book, time-pressured examination. Academic citations are never required. Discursive answers are marked on depth of explanation and application to the specific scenario, not generic textbook recall.

Question Overview

Question	Scenario	Topics	Marks
Question 1	Grow Ltd	Cash budgeting (receipts and payments), budget commentary, incremental vs rolling budgets	50
Question 2	Digi Ltd	Flexible budgeting, flexible budget and volume variances, performance evaluation, advantages of standard costing	50
Question 3	Jones Ltd	Limiting factor (key factor) analysis, production planning under a resource constraint, relevant costing, joint products/sunk/opportunity cost	50

Candidates answer any TWO of the three questions above; each is worth 50 marks. This marking scheme covers all three questions in full.

QUESTION 1**Grow Ltd**

Total: 50 marks

Grow Ltd sells organic garden fertilisers to retailers across Ireland. The Managing Director, Joy Brown, wants to ensure sufficient cash to survive the coming months. Candidates are required to (a) prepare a cash budget for Quarter Four, (b) comment on the resulting cash position with appropriate advice, and (c) differentiate between incremental and rolling budgets.

Part	Requirement	Marks
(a)	Cash budget for Q4 (Oct, Nov, Dec), showing monthly and quarterly cash position	30
(b)	Commentary on the cash budget, with advice to Joy	10
(c)	Differentiate between incremental budgets and rolling budgets	10

Question 1(a)**30 marks**

Grow Ltd, budgeted figures July–December (€): Purchases, Sales, Salaries, Overheads, and Advertising given monthly. Cash sales are 40% of total sales (5% discount applied); credit sales (60%) settled 50% in month of sale, 30% one month after, 20% two months after. Credit purchases are 50% of total purchases, paid 70% one month after purchase, 20% two months after, 10% three months after. Salaries paid one month in arrears; advertising paid two months in arrears. Overheads (including a weekly depreciation charge of €300, four weeks per month) paid one month in arrears. A new delivery van costing €60,000 is bought in October, paid in four equal monthly instalments beginning the month after purchase; the old van is sold in December for €500. Tax payments of €4,000 (September) and €15,400 (October) are expected. A €1,250 grant is expected in November. The bank balance at 30 September is expected to be €7,100. (a) Prepare a cash budget for Quarter Four, showing the firm's cash position at the end of each month AND for the quarter as a whole.

Full Model Answer – Working 1: cash receipts from sales**Cash receipts from sales**

Item	October (€)	November (€)	December (€)
Cash sales this month, net of 5% discount (40% of month's sales × 95%)	76,380	66,120	60,420
Credit sales settled in month of sale (60% of month's sales × 50%)	60,300	52,200	47,700
Credit sales settled one month after sale (prior month's credit sales × 30%)	64,080	36,180	31,320
Credit sales settled two months after sale (two months prior credit sales × 20%)	32,880	42,720	24,120
Total cash receipts from sales	233,640	197,220	163,560

Working detail (October shown): October cash sales: $40\% \times €201,000 = €80,400$, less 5% discount = €76,380. October credit sales settled this month: $60\% \times €201,000 \times 50\% = €60,300$. Settled one month after (from September's credit sales): $60\% \times €356,000 \times 30\% = €64,080$. Settled two months after (from August's credit sales): $60\% \times €274,000 \times 20\% = €32,880$.

Full Model Answer (continued) – Working 2: cash payments**Cash payments for purchases**

Item	October (€)	November (€)	December (€)
Cash purchases this month (50% of month's purchases, paid same month)	45,500	43,000	39,500
Credit purchases paid one month after purchase (prior month's credit purchases × 70%)	50,750	31,850	30,100
Credit purchases paid two months after (× 20%)	12,900	14,500	9,100
Credit purchases paid three months after (× 10%)	5,100	6,450	7,250
Total cash payments for purchases	114,250	95,800	85,950

Assumption stated: Cash purchases (50% of the total) are assumed paid in the month of purchase; only credit purchases (the other 50%) follow the stated 70%/20%/10% deferred payment pattern, since the question specifies these percentages apply only to credit purchases.

Salaries, advertising and overheads (payment timing)

Item	October (€)	November (€)	December (€)
Salaries (one month in arrears: Sep, Oct, Nov paid)	41,000	43,000	40,000
Advertising (two months in arrears: Aug, Sep, Oct paid)	14,000	15,600	10,000
Overheads, excl. €1,200 monthly depreciation, one month in arrears (Sep, Oct, Nov generated)	38,750	45,200	43,850

Overheads note: Weekly depreciation of €300 × 4 weeks = €1,200 per month is a non-cash charge and is stripped out of the overheads figure before applying the one-month payment lag: September overheads of €39,950 generate a cash payment of €39,950 – €1,200 = €38,750, paid in October, and so on.

Full Model Answer (continued) – Other items and full cash budget**Working – other cash items**

Item	October (€)	November (€)	December (€)
New van instalment (€60,000 ÷ 4, starting the month after October's purchase)	0	15,000	15,000
Sale of old van (cash inflow)	0	0	500
Tax payment	15,400	0	0
Grant received (cash inflow)	0	1,250	0

Van instalment timing: The van is bought in October but, per the question, payments are spread over four equal monthly instalments *beginning the month after purchase* — so the first instalment falls in November, not October.

Full cash budget, Quarter Four (October, November, December)

Item	October (€)	November (€)	December (€)	Quarter total (€)
Receipts				
Cash receipts from sales	233,640	197,220	163,560	594,420
Grant received	0	1,250	0	1,250
Sale of old van	0	0	500	500
Total receipts	233,640	198,470	164,060	596,170
Payments				
Payments for purchases	114,250	95,800	85,950	296,000
Salaries	41,000	43,000	40,000	124,000
Overheads (excl. depreciation)	38,750	45,200	43,850	127,800
Advertising	14,000	15,600	10,000	39,600
Tax payment	15,400	0	0	15,400
Van instalments	0	15,000	15,000	30,000
Total payments	223,400	214,600	194,800	632,800
Net cash flow for the month	10,240	(16,130)	(30,740)	(36,630)
Opening cash balance	7,100	17,340	1,210	7,100
Closing cash balance	17,340	1,210	(29,530)	(29,530)

Reconciliation: The quarter-as-a-whole net cash flow of €(36,630) equals the sum of the three monthly net cash flows (€10,240 – €16,130 – €30,740), and the closing balance for the quarter (€(29,530)) equals the opening balance at 1 October (€7,100) plus this net movement, matching the final closing balance at the end of December.

Mark Allocation

Marks	Criteria
26-30	All receipts and payments correctly calculated and timed (including the depreciation adjustment and van instalment window); monthly and quarterly closing balances all correct and reconciling.
19-25	Substantially correct with minor timing or computational errors in one or two items.
10-18	Core structure of the cash budget understood; several timing errors (e.g. cash purchases not assumed paid in month of purchase, or credit sales/purchases percentages misapplied).
0-9	Limited understanding of cash budgeting; minimal correct workings.

Question 1(b)**10 marks**

Comment upon the cash budget that you have produced in part (a), offering whatever advice to Joy that you consider appropriate.

Full Model Answer**Overall assessment**

The cash budget reveals a rapidly deteriorating cash position: Grow Ltd moves from a comfortable €17,340 balance at the end of October to an overdraft of €29,530 by the end of December — a swing of almost €47,000 in just two months, and the business is already close to the minimum viable balance by the end of November (€1,210).

What is driving the deterioration

Sales fall sharply and consistently through Q4 (from €356,000 in September to €159,000 in December), steadily reducing cash receipts, while payments remain comparatively high due to the payment-timing lags on purchases, salaries, and overheads. This is compounded by the October tax payment of €15,400 and the new €15,000-per-month van instalments beginning in November, both landing precisely as sales revenue is falling away.

Advice to Joy

1. Arrange an overdraft facility or short-term financing with the bank *in advance* of the projected December shortfall, given the budget provides clear early warning. **2.** Investigate the cause of the steep sales decline into Q4 and consider seasonal promotional activity to soften it. **3.** Review whether the timing of the new van's instalment schedule could be renegotiated to reduce the cash strain in November and December specifically. **4.** Tighten credit control on the 20% of credit sales collected two months after sale, since accelerating this collection would materially improve near-term cash receipts. **5.** Continue to monitor and update the cash budget monthly so early warning signs are acted on promptly.

Mark Allocation

Marks	Criteria
9-10	Correctly identifies the deteriorating trend and the overdraft position by December; explains the key drivers (declining sales, tax payment, van instalments); gives specific, actionable advice.
6-8	Correctly identifies the overdraft position with reasonable explanation of drivers; advice more generic.
3-5	Some relevant commentary but lacks specific reference to the figures produced in part (a).
0-2	Minimal or no meaningful commentary.

Question 1(c)**10 marks**

Differentiate between "incremental budgets" and "rolling budgets".

Full Model Answer**Incremental budget**

An incremental budget is prepared by taking the prior period's budget (or actual results) as a starting base and adjusting it incrementally for anticipated changes, such as expected inflation or planned growth, rather than building the budget from zero each period. It is quick and straightforward to prepare, but risks simply carrying forward any inefficiencies or budgetary slack built into the base period, without requiring managers to justify each line of spend afresh.

Rolling (continuous) budget

A rolling budget is continuously updated by adding a new period (e.g. a further month or quarter) onto the end of the budget as each period elapses and is dropped off, so the organisation always maintains a budget covering a full period ahead. This requires more frequent preparation and administrative effort, but keeps the budget more current and responsive to changing conditions — particularly valuable for a business like Grow Ltd facing sharply declining seasonal sales, where a cash budget prepared once and left unrevised would quickly become an unreliable guide to the business's actual cash needs.

Key distinction: The incremental/rolling distinction concerns how the *content* of a budget is derived (from a prior base, adjusted incrementally, vs continuously extended), which is a separate concept from a fixed vs a flexible budget (the distinction tested in Question 2), which concerns how a budget is *compared against actual results* after the fact.

Mark Allocation

Marks	Criteria
8-10	Both terms accurately and fully defined, with a clear explanation of the trade-off (speed/simplicity vs currency/responsiveness) between them, ideally applied to Grow Ltd's context.
5-7	Both terms defined with reasonable accuracy; trade-off or application less developed.
2-4	One term defined well; the other only partially, or the distinction between them unclear.
0-1	No meaningful explanation of either term.

QUESTION 1 TOTAL — 50 MARKS

QUESTION 2**Digi Ltd**

Total: 50 marks

Digi Ltd produces the "Learn" tablet. The Managing Director, John Wise, wants to assess 2022 trading performance, comparing budget to actual, where actual production and sales were only 85% of the original 15,000-unit target due to supply chain issues. Candidates are required to (a) construct a flexible budget and calculate flexible budget variances and volume variances, (b) critically evaluate Digi Ltd's financial performance based on that analysis, and (c) outline three advantages of standard costing.

Part	Requirement	Marks
(a)	Flexible budget, flexible budget variances (FBV) and volume variances (VV), each marked favourable or unfavourable	24
(b)	Critical evaluation of 2022 financial performance and advice to Mr. Wise	14
(c)	Three advantages of standard costing	12

Question 2(a)**24 marks**

Digi Ltd, 2022, "Learn" tablet: Budget (based on 15,000 units) and Actual figures given for sales revenue, variable costs (direct materials, direct labour, production overheads, advertising, delivery) and fixed costs (insurance, rent, rates, salaries). Actual production and sales were 85% of the 15,000-unit target. (a) Construct a flexible budget for Digi Ltd for 2022. Then, using the flexible budget, master budget and actual amounts, calculate flexible budget variances (FBVs) and volume variances (VVs) for each budgetary item, indicating whether each variance should be considered favourable (F) or unfavourable (U).

Full Model Answer**Working – actual volume and per-unit budget rates**

Item	Value
Master budget volume	15,000 units
Actual volume (85% of target)	12,750 units
Budgeted selling price per unit ($€1,950,000 \div 15,000$)	€130.00

Method: The flexible budget re-states the master budget's variable items at the *actual* volume of 12,750 units, using the *budgeted* rate per unit for each; fixed costs remain unchanged in the flexible budget. The flexible budget variance (FBV) isolates genuine spending/price/efficiency differences at the same, actual volume; the volume variance (VV) isolates the effect of the volume shortfall alone, at budgeted rates.

Full Model Answer (continued) – Flexible budget and variances: variable items**Flexible budget, FBV and VV – sales revenue and variable costs**

Item	Master budget (€)	Flexible budget (€)	Actual (€)	FBV (Actual – Flex)	VV (Flex – Master)
Sales revenue	1,950,000	1,657,500	1,593,750	(63,750) U	(292,500) U
Direct materials	825,000	701,250	739,500	(38,250) U	(123,750) F
Direct labour	375,000	318,750	350,625	(31,875) U	(56,250) F
Production overheads	180,000	153,000	140,250	12,750 F	(27,000) F
Advertising	45,000	38,250	31,875	6,375 F	(6,750) F
Delivery	22,500	19,125	22,950	(3,825) U	(3,375) F

Sign convention for costs vs revenue: For sales revenue, a shortfall against the comparison figure is unfavourable. For costs, a **lower** actual/flexible figure is favourable (less was spent than expected), even though the raw arithmetic (Actual – Flex, or Flex – Master) may be negative — each variance is labelled F or U according to its economic meaning, not the arithmetic sign.

Contribution

Item	Master budget (€)	Flexible budget (€)	Actual (€)	FBV	VV
Contribution	502,500	427,125	308,550	(118,575) U	(75,375) U

Full Model Answer (continued) – Fixed costs and net profit**Flexible budget, FBV and VV – fixed costs**

Item	Master budget (€)	Flexible budget (€)	Actual (€)	FBV (Actual – Flex)	VV (Flex – Master)
Insurance	60,000	60,000	72,000	(12,000) U	0
Rent	80,000	80,000	85,000	(5,000) U	0
Rates	25,000	25,000	22,000	3,000 F	0
Salaries	150,000	150,000	170,000	(20,000) U	0
Total fixed costs	315,000	315,000	349,000	(34,000) U	0

Why fixed costs have no volume variance: Fixed costs are, by definition, unaffected by the volume of activity within the relevant range, so the flexible budget figure for every fixed cost line equals the master budget figure — the volume variance for every fixed cost item is therefore zero.

Net profit

Item	Master budget (€)	Flexible budget (€)	Actual (€)	FBV	VV
Net profit (loss)	187,500	112,125	(40,450)	(152,575) U	(75,375) U

Overall reconciliation: The total variance in net profit (Actual €(40,450) – Master budget €187,500 = €(227,950) unfavourable) is fully explained by the sum of the flexible budget variance (€(152,575) unfavourable) and the volume variance (€(75,375) unfavourable): €(152,575) + €(75,375) = €(227,950). This is the most severe unfavourable performance seen across this style of question in the series, with the master budget's projected €187,500 *profit* turning into an actual *loss* of €40,450.

Mark Allocation

Marks	Criteria
20–24	Actual volume correctly derived (12,750 units); flexible budget correctly constructed for all variable items at budgeted per-unit rates; fixed costs correctly held constant; all FBVs and VVs correctly calculated and correctly labelled F/U; net profit reconciliation correct, including the budget-to-loss turnaround.
14–19	Substantially correct with minor errors in one or two line items, or F/U labelling inconsistently applied.
7–13	Flexible budget concept understood but significant errors in the FBV/VV split, or fixed costs incorrectly flexed with volume.
0–6	Limited understanding of flexible budgeting; minimal correct workings.

Question 2(b)**14 marks**

Based upon your answer to (a), critically evaluate Digi Ltd.'s financial performance during 2022, offering whatever advice to Mr. Wise that you consider appropriate.

Full Model Answer**Headline result: a budgeted profit turned into an actual loss**

Digi Ltd budgeted a healthy €187,500 profit for 2022, but actually recorded a €40,450 **loss** — a swing of €227,950 in the wrong direction. This is a materially worse outcome than in comparable questions elsewhere in this series and warrants serious attention from Mr. Wise.

The volume variance (€(75,375) unfavourable) reflects the supply chain disruption

This portion of the shortfall stems directly from the 15% fall in production and sales volume caused by supply chain issues. While concerning in its own right (it points to a vulnerability in Digi Ltd's supply arrangements), this component does not, on its own, reflect poor day-to-day operational or cost management during the period.

The flexible budget variance (€(152,575) unfavourable) is the dominant and most concerning factor

This is double the volume variance and represents genuine spending and efficiency problems *after* already adjusting for the lower volume achieved. Direct materials and direct labour are both significantly unfavourable at the actual volume (€(38,250) and €(31,875) respectively) — unusual, since with lower volume one might expect easier cost control, suggesting either input price inflation or productivity/efficiency problems on the shop floor. On fixed costs, **salaries** (€(20,000) unfavourable) and **insurance** (€(12,000) unfavourable) are the standout overruns.

Advice to Mr. Wise

Investigate direct material and labour cost overruns as the priority, since these represent genuine inefficiency or price increases even after allowing for lower volume — possible causes include raw material price rises, overtime working, or reduced labour productivity. Investigate the salary and insurance increases separately (unbudgeted pay rises, new hires, or a premium increase). Given the scale of the shortfall, Digi Ltd should also urgently review the resilience of its supply chain, since a further disruption of similar magnitude next year could threaten the business's viability if cost control is not also addressed.

Mark Allocation

Marks	Criteria
12-14	Correctly separates the volume-driven shortfall from the (larger, more serious) flexible budget variance; correctly identifies the unusual unfavourable material/labour variances alongside salary/insurance overruns; gives specific, evidence-based, appropriately serious advice given the budget-to-loss turnaround.
8-11	Good evaluation distinguishing FBV from VV with reasonable identification of key drivers; advice more generic.
4-7	Some relevant commentary but does not clearly distinguish the two variance types, or lacks specific reference to individual line items.
0-3	Minimal or no meaningful evaluation.

Question 2(c)**12 marks**

Outline and briefly explain THREE advantages associated with the use of standard costing.

Full Model Answer**1. Facilitates budgeting, planning and variance analysis**

Standard costs provide a pre-determined, reliable cost per unit for materials, labour and overheads, which forms the basis for setting budgets before a period even begins (as with Digi Ltd's 15,000-unit master budget). This same standard cost data directly enables the flexible budgeting and variance analysis performed in part (a) — without agreed standard costs per unit, it would not be possible to construct a flexible budget or meaningfully isolate flexible budget variances from volume variances.

2. Supports management by exception

By comparing actual results against pre-set standards, management's attention is naturally directed towards the areas showing the largest variances (such as Digi Ltd's material and labour cost overruns identified in part (b)), rather than requiring every transaction or cost line to be reviewed in detail. This makes performance monitoring far more efficient, allowing scarce management time to be focused where it is most needed.

3. Provides a benchmark for performance evaluation and control

Standard costs create a clear, objective benchmark against which the efficiency of departments, processes, and individual managers can be assessed, supporting responsibility accounting and, potentially, performance-related incentive schemes. This can motivate staff and managers towards cost control and operational efficiency, since performance against a known standard is more meaningful than performance against last year's actual results alone.

Other acceptable advantages: Candidates may alternatively discuss: standard costing simplifies inventory valuation and cost record-keeping, since units can be valued at a consistent standard cost rather than tracking fluctuating actual costs unit by unit; or that standard costs provide a useful starting point for price-setting decisions. Any three well-explained, distinct advantages receive full credit.

Mark Allocation

Marks	Criteria
10-12	Three distinct, well-explained advantages given, each a genuine and recognised benefit of standard costing, ideally with at least one linked to the flexible budgeting/variance work in parts (a) and (b).
7-9	Three advantages identified with reasonable explanation, or two explained in excellent depth.
4-6	Two advantages identified with limited explanation.
0-3	One or no relevant advantages identified.

QUESTION 2 TOTAL — 50 MARKS

QUESTION 3**Jones Ltd**

Total: 50 marks

Jones Ltd produces four types of steel knives ("A", "B", "C" and "D") and faces an anticipated global shortage of steel in 2024, limiting total supply to 50,000 kilograms. Candidates are required to (a) formulate a profit-maximising production plan under this constraint, (b) outline four actions to address the demand shortfall, (c) outline the three criteria for a relevant cost, and (d) define three key terms.

Part	Requirement	Marks
(a)	Profit-maximising production plan under the steel constraint, with assumptions and commentary	20
(b)	Four potential actions to meet total demand in 2024	12
(c)	Three criteria for a relevant cost/benefit	9
(d)	Define joint products, sunk cost, and opportunity cost	9

Question 3(a)**20 marks**

Jones Ltd, 2024 budgeted unit information for four knives (A, B, C, D): sales price, direct material costs (€20/kg of steel), direct labour, production overheads (50% variable), marketing costs (40% variable), other costs (60% variable), and sales demand given for each. Total steel supply is limited to 50,000 kilograms. (a) Assuming Jones Ltd. wishes to maximise profits in 2024, formulate a production plan for the firm subject to the constraints faced by them. State any assumptions made and comment briefly upon the production plan formulated.

Full Model Answer**Step 1 - confirm steel is a genuine limiting factor**

Kilograms of steel required per unit = direct material cost per unit ÷ €20 per kilogram. Total kilograms required to meet full budgeted demand for all four products is calculated below and compared against the 50,000 kg supply.

Working - kilograms required per unit and to meet full demand

Product	DM cost/unit (€)	Kg/unit (+€20)	Sales demand (units)	Total kg needed
A	50.00	2.500	10,000	25,000
B	80.00	4.000	6,000	24,000
C	100.00	5.000	3,000	15,000
D	130.00	6.500	1,000	6,500
Total kg needed for full demand				70,500

Constraint confirmed: 70,500 kilograms would be needed to meet full demand for all four products, against a supply of only 50,000 kilograms. Steel is therefore a genuine limiting (key) factor, and production must be prioritised across products to maximise profit from the available supply.

Full Model Answer (continued) – Step 2: contribution per unit of limiting factor

Direct materials and direct labour are wholly variable. Half of production overheads are variable; 40% of marketing costs are variable; 60% of other costs are variable. Variable cost per unit therefore = DM + DL + 50% of production overheads + 40% of marketing + 60% of other costs.

Working – variable cost and contribution per unit

Product	A	B	C	D
Sales price	150.00	180.00	240.00	270.00
Direct materials	50.00	80.00	100.00	130.00
Direct labour	20.00	24.00	28.00	32.00
Variable production overheads (50%)	7.50	11.00	13.00	16.00
Variable marketing costs (40%)	4.80	5.60	8.00	11.20
Variable other costs (60%)	14.40	18.00	20.40	27.60
Total variable cost	96.70	138.60	169.40	216.80
Contribution per unit	53.30	41.40	70.60	53.20

Working – contribution per kilogram of steel (the limiting factor), and ranking

Product	Contribution/unit (€)	Kg/unit	Contribution/kg (€)	Rank
A	53.30	2.500	21.32	1st
C	70.60	5.000	14.12	2nd
B	41.40	4.000	10.35	3rd
D	53.20	6.500	8.18	4th

Key point on ranking: Ranking products by contribution *per unit* alone would give the wrong answer here: Product C has the highest contribution per unit (€70.60), and Product A ranks only third by that measure (€53.30, barely ahead of D's €53.20). But once contribution is expressed per unit of the scarce resource (steel), Product A jumps to the clear top rank (€21.32 per kg) and Product D falls to last place (€8.18 per kg) — the correct basis for prioritising production under a resource constraint is always contribution per unit of the limiting factor.

Full Model Answer (continued) – Step 3: allocate the 50,000 kg and finalise the production plan**Working – allocation of available steel (50,000 kg) in rank order**

Rank	Product	Kg needed for full demand	Kg allocated	Cumulative kg used	Units produced
1st	A	25,000	25,000	25,000	10,000 (full demand)
2nd	C	15,000	15,000	40,000	3,000 (full demand)
3rd	B	24,000	10,000 (remaining balance)	50,000	2,500 (partial)
4th	D	6,500	0	50,000	0

Product B partial production: After meeting full demand for A (25,000 kg) and C (15,000 kg), only $50,000 - 40,000 = 10,000$ kg remain. At 4.000 kg per unit, this supports $10,000 \div 4.000 = 2,500$ units of B, out of the 6,000 units demanded — leaving a shortfall of 3,500 B units, and no capacity remaining at all for Product D.

Resulting profit-maximising production plan and contribution

Product	Units to produce	Contribution/unit (€)	Total contribution (€)
A	10,000	53.30	533,000
B	2,500	41.40	103,500
C	3,000	70.60	211,800
D	0	53.20	0
Total contribution			848,300

Assumptions made

1. Fixed costs (the fixed portions of production overheads, marketing, and other costs) are assumed to remain unchanged regardless of the production mix, so maximising total contribution is equivalent to maximising total profit. **2.** Steel is assumed to be the only binding constraint on production — other resources (labour, machine capacity) are assumed sufficient to produce whatever mix the steel supply allows. **3.** Products are assumed divisible for the purposes of fully utilising the scarce resource; in practice, Product B production would need to be a whole number of saleable units (2,500 exactly, as it happens, requires no rounding here).

Brief commentary on the plan

The optimal plan fully satisfies demand for the two most steel-efficient products (A and C) but leaves a significant 3,500-unit shortfall in B and eliminates D production entirely, despite D being Jones Ltd's highest-priced knife (€270). Kate Gold should recognise the plan deliberately allocates the steel shortage away from the least resource-efficient products (measured per kilogram, not per unit), which may need to be managed carefully from a customer relationship and premium-range positioning perspective, even though it is the profit-maximising response to the constraint as it currently stands.

Mark Allocation

Marks	Criteria
17-20	Limiting factor correctly confirmed; variable cost, contribution per unit, and contribution per kg all correctly calculated; correct ranking (crucially, correctly re-ranking A ahead of C despite C's higher per-unit contribution); correct allocation of the 50,000 kg giving the profit-maximising plan; clear assumptions and relevant commentary.
12-16	Substantially correct approach with a minor computational error, e.g. in one variable cost split.
6-11	Limiting factor approach understood in principle but ranked by contribution per unit rather than per unit of limiting factor, or allocation contains significant errors.
0-5	Limited understanding of limiting factor analysis; minimal correct workings.

Question 3(b)**12 marks**

Outline and briefly explain FOUR actions that could potentially be undertaken by Jones Ltd. in an attempt to ensure that they are able to satisfy the total demand for their knives in 2024.

Full Model Answer**1. Source alternative or additional steel suppliers**

Jones Ltd could seek out additional suppliers of steel, potentially in different geographic regions less affected by the anticipated global shortage, to increase total available supply above 50,000 kilograms.

2. Redesign products to reduce steel usage

Investing in product redesign or engineering work to reduce the quantity of steel required per knife (without significantly compromising quality, durability, or the 'award-winning' reputation of the range) would increase the number of units that can be produced from the same total supply.

3. Secure long-term or advance supply agreements

Entering into long-term contracts or advance purchase agreements with existing steel suppliers could help secure priority access to available supply ahead of competitors also facing the same global shortage, reducing the risk of further supply disruption.

4. Identify substitute materials

Jones Ltd could research and qualify substitute materials or alloys that could partially replace scarce steel in some knife designs, freeing up the remaining steel supply for products where substitution is not feasible without compromising quality.

Other acceptable actions: Candidates may alternatively discuss: raising prices on the most steel-intensive products to manage/ration demand down towards available supply; building buffer stock of steel in advance where feasible; or outsourcing production of some knives to a third party with independent access to steel supply. Any four well-explained, distinct actions receive full credit.

Mark Allocation

Marks	Criteria
10-12	Four distinct, well-explained actions given, each a genuine and practical response to a raw-material supply constraint.
7-9	Four actions identified with reasonable explanation, or three actions explained in excellent depth.
4-6	Two to three actions identified with limited explanation.
0-3	One or no relevant actions identified.

Question 3(c)**9 marks**

Outline and briefly explain the THREE criteria that must be met for any cost (or benefit) to be considered as relevant for decision-making purposes.

Full Model Answer**1. It must be a future cost or benefit**

A relevant cost or benefit must relate to something that will occur in the future, as a consequence of the decision being made now. Costs already incurred (sunk costs, see part (d)) cannot be changed by the decision and are therefore never relevant.

2. It must differ between the alternatives being considered (it must be incremental/differential)

A cost or benefit is only relevant if it changes depending on which course of action is chosen. Any cost or benefit that is identical regardless of which alternative is selected has no bearing on the decision and can be ignored, even if it is a genuine future cash flow.

3. It must represent an actual cash flow

Relevant costs and benefits must correspond to real cash inflows or outflows. Purely accounting entries or allocations — such as depreciation charges, or absorbed fixed overhead that would be incurred regardless of the decision — do not represent genuine cash flows arising from the decision and are not relevant, even though they may appear in the accounting records.

Mark Allocation

Marks	Criteria
8-9	All three criteria correctly identified and clearly explained, using accurate terminology (future, incremental/differential, cash flow).
5-7	All three criteria identified with reasonable explanation, or two explained in excellent depth.
2-4	One or two criteria identified with limited explanation.
0-1	No relevant criteria identified.

Question 3(d)**9 marks**

Outline and briefly explain what is meant by each of the following terms: (i) Joint products; (ii) Sunk cost; (iii) Opportunity cost.

Full Model Answer**(i) Joint products**

Joint products are two or more distinct products that are produced simultaneously from the same input, process, or raw material, up to a 'split-off point' beyond which they become separately identifiable and may require further individual processing. An example would be a single batch of steel processing yielding offcuts or by-products usable across Jones Ltd's different knife ranges.

(ii) Sunk cost

A sunk cost is a cost that has already been incurred in the past and cannot be recovered, avoided, or changed by any current or future decision. Because it does not differ between the alternatives being considered, a sunk cost is irrelevant to decision-making and should be ignored.

(iii) Opportunity cost

An opportunity cost is the value of the benefit forgone by choosing one course of action over the next best available alternative. It does not appear anywhere in the accounting records, but represents a genuine economic sacrifice and is therefore highly relevant to decision-making — for example, the contribution Jones Ltd forgoes by using its limited steel supply on Product B rather than Product D is an opportunity cost of the production plan chosen in part (a).

Mark Allocation

Marks	Criteria
8-9	All three terms accurately and fully defined, with a relevant example given for at least one, ideally tied to the Jones Ltd scenario.
5-7	All three terms defined with reasonable accuracy; examples missing or generic.
2-4	One or two terms defined accurately.
0-1	No meaningful definitions provided.

QUESTION 3 TOTAL — 50 MARKS